

A Multi-Stage Deep Learning Framework for Smartphone E-Waste Classification, Damage Assessment, Metal Value Estimation, and Recycling Recommendation

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Abstract—Smartphone e-waste is a growing waste stream with “reusable” smart phones, “recoverable” metals and “environmentally sensitive” materials. The handlings should not only involve a determination of a category of the device but also a determination of the model, determination of the condition, determination of the severity of the damage, determination of the value of the material and a final determination for resale, repair or recycling. In this paper, a deep-learning-based decision-support framework is presented for assessing e-waste present in a smartphone image from a single image input in five stages. Stage-A has determined the smartphone model to be among the iPhone 8, iPhone X, iPhone 11 and iPhone 13. Stage-B: Normal or broken. Stage-C is conditionally executed if the device is broken and the damage severity is minor or major as predicted by Stage-B. Stage-D calculates the value of metal that can be recycled in Bangladeshi taka based on a model-specific database of metal compositions which may be MetalpriceAPI live or delayed pricing or CSV pricing. Stage-E uses deterministic rule based logic to suggest ‘as is’, ‘repair as is’, ‘recycle’ and/or ‘assess manually’. Three model families were tested throughout the classification stages: CNN baseline DenseNet121, practical comparison model YOLO11x-cls and the final model family chosen as the pipeline: Vision Transformer-based models. The final models reached 91.91% test accuracy on Stage-A with DINOv2 ViT-B/14, 92.12% with Swin Transformer V2 Tiny and 94.67% with MaxViT-Tiny on Stage-C. The entire framework was incorporated into a web demonstration system built with FastAPI and React that displayed a prototype for an image-based decision-support system for e-waste management in a smartphone.

Index Terms—Smartphone, e-waste, deep learning, Vision Transformer, DINOv2, Swin Transformer V2, MaxViT, YOLO11x-cls, DenseNet121, image classification, damage assessment, metal value estimation, recycling recommendation.

I. INTRODUCTION

The problem of e-waste has arisen with the surge in consumer electronics and the relatively short life span for new gadgets. In this regard, smart phones play a particularly important role, as the device is rich in valuable metals, reusable parts, plastics, glass and materials related to the battery. A discarded smartphone can still be found useful for resale, repair, refurbishment, recovery of spare parts or material recycling. The most suitable route chosen will be based on the model of device, visibility, severity of damage and estimated material value of device recovery.

Assessment of e-waste is typically done manually in the traditional way. The device is inspected visually by a technician or recycler, the model is identified, and the device is determined to be damaged, the degree of damage is estimated, and then the decision is made whether the device is to be resold, repaired or recycled. This process is tedious, subjective and un-scalable for large amounts of discarded phones. A computer vision-based system can assist this process by generating homogeneous initial forecasts from smartphone images and final decisions can still be left to human supervision.

In most image classification systems, only one label is produced, for example that of the category of the object. But the e-waste management of smartphones is indeed a multi-stage decision problem. A helpful system ought to not just match the model of the device but additionally assess the situation, determine damage state if required, approximate the worth of the materials, and provide sensible advice for dealing with the gadget. Thus, this study suggests an incremental approach rather than a single big classifier.

The salient features of this work are as follows: A five-stage pipeline for e-waste decision support for smartphone model classification, condition classification, damage severity classification, metal value estimation and recommendation generation is developed. Second, a custom smartphone e-waste data set is gathered from online sources and real world images from phone repair and e-waste shops. Third, the model families of CNN, YOLO and Vision Transformer are evaluated at three different learning-based classification phases. Finally, the last deployed pipeline is made up of DINOv2 ViT-B/14, Swin Transformer V2 Tiny, and MaxViT-Tiny. Fifth, it integrates a hybrid metal value estimation module with MetalpriceAPI and fallback pricing. Lastly, the entire structure is put into practice as a web demonstration system using FastAPI/React.

II. RELATED WORK

E-waste management is a multi-disciplinary issue related to environment, resource recovery, policy, logistics and technology. The importance of proper handling of e-waste from economic and environmental perspectives has been conveyed in the previous studies as the materials contained in waste electronics have both value and hazardous materials [1]–[4]. Smartphone recycling is of special significance since the devices have recoverable materials like gold, silver, copper, aluminium, cobalt, titanium, rare earth elements, etc. However, MRV is affected by collection, sorting, condition of the devices, recovery rate, and market price.

There have been many successful applications of computer vision in waste classification, smart-bin systems, sorting of recyclable material, and object recognition. The general waste datasets and the classifiers based on Convolutional Neural Networks have demonstrated the potential of visual classification to aid automated sorting [5] [6]. Recent research in the field of e-waste includes deep learning for e-waste image classification [7] which has been done to explore more recent research work on e-waste specific. Many current systems, however, only go so far as a single category label and fail to link the visual prediction with downstream decision making (e.g., resale, repair or recycling).

CNN architectures like ResNet, DenseNet, MobileNet, and EfficientNet have been popular in the visual recognition field [8]. DenseNet is helpful as a baseline since its dense connections promote feature reuse and parameter efficiency [8]. While object detection is a common application of YOLO models, modern versions of Ultralytics have extended their capabilities to include image classification, making YOLO11x-cs an appropriate benchmark model for deployment-focused image classification [9].

Vision Transformers treat the image as a sequence of patches, and apply self-attention to capture global relationships [10]. This can be beneficial for smartphone model recognition, which requires the system to identify the overall visual features, like the camera’s layout, the shape of the smartphone, the presence of holes or a notch, and the design of the frame. To improve the visual representation, hierarchical transformer models such as Swin Transformer and Swin Transformer

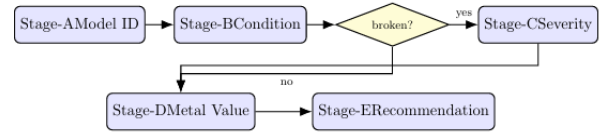


Fig. 1. Five-stage smartphone e-waste decision-support pipeline.

V2 employ shifted-window attention and hierarchical feature extraction using scalable windows [11], [12]. The multi-axis attention enables MaxViT to focus on both local and global information, which allows it to be applied to tasks involving both the local damage pattern and the global structural context [13]. DINOv2 is a powerful self-supervised visual feature extractor that is applicable to many image domains [14].

This paper has identified the research gap in e-waste assessment pipeline for smart phone, which is the absence of multi-stage image classification, value estimation based on the live or delayed metal prices and recommending recycling based on rules in a single prototype.

III. METHODOLOGY

A. Overall Pipeline Design

The following are five stages proposed in the framework. Stage-A does smartphone model classification and predicts one of four classes: iPhone 8, iPhone X, iPhone 11 or iPhone 13. Stage-B makes a binary classification of conditions and forecasts normal or broken. Stage-C is run if Stage-B thinks the device is broken, and it does damage severity classification. The minor damage and major damage predictions are those of Stage-C. The value of recyclable metal in BDT is estimated with the use of the stage-A predicted model and the metal composition database named Stage-D. The final recommendation is generated by Stage-E, which is based on rule-based decision logic.

It is important that the conditional execution of stage-c is in place because normal phones need not be labeled as minor or major damage. This design also lends interpretability since each stage of the design asks a different question in the recycling decision process.

B. Dataset Construction

The custom dataset was prepared because a dataset that fits all these needs and contains the selected iPhone models was not found. A custom dataset was created, since no publicly available dataset was found which could be used for all these needs, including the smartphone model classification, normal/broken classification and minor/major damage classification for the selected iPhone models. Images were retrieved from public sources available online including Google Images, Facebook Marketplace, Pinterest, product review sites, resale sites and ecommerce sites. Further realistic images were obtained by visiting local phone repair and scrap/vangari stores in order to view the realistic damage patterns.

The data set was split into three separate stage-wise data sets. Stage-A has 4,000 images from 4 model classes of

TABLE I
STAGE-WISE DATASET COMPOSITION

Stage	Task	Train	Validation	Test	Total	No. of Classes
Stage-A	Smartphone model classification	3,036	482	482	4,000	4
Stage-B	Normal vs. broken condition classification	7,036	482	482	8,000	2
Stage-C	Minor vs. major damage severity classification	2,197	450	450	3,097	2

iPhones. There are 8,000 images in stage-B and two classes of conditions. Stage-C has 3,097 images from two damage severity classes in it. The data sets were divided into the three sets (training, validation and test sets) without overlap during training.

C. Label Definitions

Model labels on the smartphones were determined visually. iPhone 8 was recognised as the rear camera and the physical home button. iPhone X is distinguished by the Face ID notch and vertical dual rear camera. iPhone 11 was spotted with a Face ID notch and dual cameras in a square camera block. The smaller notch and diagonal camera setup were the tell-tale signs of the iPhone 13.

Normal class consists of shots where no cracks in the screen, major damage to the frames, broken glass or significant body damage can be seen. Images showing cracks, broken glass, missing components, damage to the frame, bending, or broken displays will be classified as broken images. Minor damage refers to damage that is not extensive or widespread, and includes the following: small cracks, corner cracks, surface scratches, and small chips. Major damage is defined as significant visible damage, cracking across the entire screen, loss of glass, complete separation of the screen, severe deformation or multiple structural failures.

D. Preprocessing and Augmentation

All images were checked, converted to RGB images, resized to model size, converted to tensors, and normalized with the mean and standard deviation of the ImageNet database. The input size is 224×224 for DinoV2, MaxViT, and DenseNet, and is 256×256 for the model configuration of Swin Transformer V2 Tiny. Only training images were subjected to augmentation. The training augmentation pipeline contained random horizontal flipping, random rotation, random resized cropping, color jitter, affine transformation, Gaussian blur, random grayscaling, and random erasing. To ensure the evaluation was reproducible, validation and test sets were not augmented.

E. Model Families

A pilot study was conducted on three model families. The primary CNN baseline was DenseNet121. For practical classification comparison, YOLO11x-cls model was used. The final pipeline that was selected was based on a Vision Transformer-based model. DINOv2 ViT-B/14 is used in Stage-A, Swin Transformer V2 Tiny in Stage-B, MaxViT-Tiny in Stage-C.

The pretrained weights from each model were loaded for all models and fine-tuned on their stage-wise datasets. DINOv2 followed the procedure of attaching a classification head to the CLS token representation. Swin Transformer V2 Tiny and

TABLE II
MODEL FAMILIES USED IN THE STUDY

Model Family	Model	Pretraining Source	Purpose
CNN	DenseNet121	ImageNet-1K	Baseline comparison
YOLO	YOLO11x-cls	ImageNet-1K	Practical classification comparison
Vision Transformer	DINOv2 ViT-B/14	LVD-142M	Final Stage-A model classification
Vision Transformer	Swin Transformer V2 Tiny	ImageNet-1K	Final Stage-B condition classification
Vision Transformer	MaxViT-Tiny	ImageNet-1K	Final Stage-C severity classification

TABLE III
FIVE-STAGE PIPELINE SUMMARY

Stage	Task	Input	Output	Execution Condition
Stage-A	Smartphone model classification	Smartphone image	iPhone 8, iPhone X, iPhone 11, or iPhone 13	Always
Stage-B	Condition classification	Smartphone image	Normal or broken	Always
Stage-C	Damage severity classification	Smartphone image	Minor damage or major damage	Only if Stage-B predicts broken
Stage-D	Metal value estimation	Stage-A predicted model	Estimated recyclable value in BDT	Always
Stage-E	Recommendation generation	Stage-B and Stage-C outputs	Resell, repair, recycle, or manual review	Always

MaxViT-Tiny were imported from timm with the classification heads replaced by the output layers of each stage. Optimized using AdamW.

F. Metal Value Estimation

Stage-D correlates the expected smartphone model to a database on metals, which includes the approximate amounts of gold, silver, copper, aluminium, cobalt, titanium and the rare earth elements that are recoverable. The recyclable value is based on the following:

$$V_{BDT} = \left(\sum a_m \times r_m \times p_m \right) \times c \quad (1)$$

where a_m is the amount of metal m which is thought to be recovered in grams, r_m is the recovery rate, p_m is the USD price per gram of the metal, and c is the USD to BDT exchange rate. The live or delayed prices of supported metals, USD-to-BDT exchange rate are retrieved using MetalpriceAPI. In the case of an API failure or unsupported metals (titanium/rare earth elements), a back-up CSV is used.

G. Recommendation Logic

The outputs of the classifications are translated into an actual decision in Stage-E. If Stage-B forecasts normal, then proceed to sell as is. If Stage-B is broken and Stage-C is minor damage then it suggests whether to repair or resell which depends on the damage severity. If Stage-B says broken, and Stage-C says major damage, then it is recommended to be recycled. In cases where prediction is not available, or is not certain, manual review is recommended.

IV. RESULTS

A. CNN Baseline Results

DenseNet121 was used as the CNN baseline. It scored 80.08% test accuracy for Stage-A, 83.40% for Stage-B, and 78.89% for Stage-C. The findings indicate that CNN baselines can learn a number of useful smartphone features, but the visual similarity of the models and damage patterns limit them to a lower performance.

TABLE IV
RULE-BASED RECOMMENDATION LOGIC

Stage-B Prediction	Stage-C Prediction	Final Recommendation
Normal	Not executed	Resell as-is
Broken	Minor damage	Repair then resell
Broken	Major damage	Recycle
Uncertain	Unavailable	Manual review required

TABLE V
FINAL SELECTED VISION TRANSFORMER MODEL RESULTS

Stage	Final Selected Model	Test Accuracy	Weighted F1-Score	Best Validation Accuracy
Stage-A	DINOv2 ViT-B/14	91.91%	91.90%	94.19%
Stage-B	Swin Transformer V2 Tiny	92.12%	92.08%	95.44%
Stage-C	MaxViT-Tiny	94.67%	94.66%	92.00%

B. YOLO11x-cls Results

YOLO11x-cls outperformed DenseNet121 at each stage. It scored 89.42% test accuracy for Stage-A, 88.38% for Stage-B, and 89.56% for Stage-C. This is to demonstrate that the YOLO classification is a competitive practical method. The final Vision Transformer models, however, were better than YOLO11x-cls in all three stages.

C. Vision Transformer Results

The best results were obtained for the final selected Vision Transformer models. For Stage-A, DINOv2 ViT-B/14 outperformed all other methods in test accuracy (91.91%) and weighted F1 (91.90%) with best validation accuracy of 94.78% and best validation F1 of 94.77%. For Stage-B, DINOv2 ViT-B/14 outperformed all other methods in test accuracy (92.12%) and weighted F1 (92.08%) with best validation accuracy of 95.44% and best validation F1 of 95.42%. The best test accuracy and test weighted F1-score for Stage-C are 94.67% and 94.66%, respectively, obtained by MaxViT-Tiny.

D. Cross-Family Comparison

The final VIT models showed better results in all the trials as compared to CNN and YOLO models. The improvement compared to the DenseNet121 CNN baseline was 11.83 percentage points for Stage-A, 8.71 percentage points for Stage-B and 15.78 percentage points for Stage-C; and compared to YOLO11x-cls, the improvement was 2.49 percentage points for Stage-A, 3.74 percentage points for Stage-B and 5.11 percentage points for Stage-C.

E. Stage-Wise Analysis

The most frequent mistake was in determining the differences between iPhone X and iPhone 11, and this is explainable since both phones have the same notch in front; they both have a rectangular body. DINOv2 self-supervised visual features were demonstrated to boost discrimination between similar models of devices.

After anti-overfitting tuning with weight decay and progressive unfreezing, Swin Transformer V2 Tiny has obtained good results in Stage-B. The highest validation accuracy was 95.44% and test accuracy was 92.12%. The best thing to watch for downstream is false negatives, a broken phone, is

mistakenly identified as normal—thus an opportunity to resell is missed.

MaxViT-Tiny gave the best stage-wise accuracy for the stage-C with a value of 94.67%. Its multi-axis attention is well adapted to capture both local damages of cracking and global damages of structures. According to the confusion matrix, 205 images of major damage were correctly classified while 20 were wrongly labeled as minor damages and 221 minor images were correctly labeled with 4 belonging to the minor class misclassified as major class.

F. Final Pipeline and Cascading Accuracy

The last five-stage pipeline fuses the three chosen Vision Transformer models with the metal value estimation in Stage-D and the recommendation logic in Stage-E. Stage-A, Stage-B and Stage-C are sequential classification stages, so errors will cascade. The accuracy of the theoretical model of the cascading process can be estimated by multiplying the three classification accuracies:

$$0.9191 \times 0.9212 \times 0.9467 = 0.80154 \approx 80.154\%$$

This value is not a measured end-to-end deployment accuracy. It is a theoretical approximation that makes the assumption that all three classification stages are necessary and errors are independent. In practice, only broken devices are used to execute Stages-C and Stage-D and Stage-E are rule-based modules.

G. MetalpriceAPI Validation

In the last successful run, the price data from the APIs were successfully retrieved for gold, silver, copper, aluminium, cobalt, and the exchange rate between the USD and BDT. Csv fallback symbols were used for Titanium and rare earth elements as they don't have a direct API symbol in the implemented pricing setup. This is a hybrid design which allows the system to operate when the API is not available, or when there is a metal missing for which no API exists.

V. DISCUSSION

The findings indicate that a multi-stage pipeline with a structure is better suited for the smartphone e-waste decision making in comparison to a single-label classifier. Every classification task is different with various visual requirements. The Stage-A is for device identity features, like camera layout and body design, which are required globally. The condition-level visual inspection for Stage-B will include cracks and any visible damage. In the case of Stage C, there is a need for a fine-grained severity analysis that takes into account both damage in the local area and damage to the structure as a whole. The experimental results, therefore, make the use of different Vision Transformer-based models for the different stages justified.

The system is also interpretable, since explicit rules are used to provide the final recommendation. This can be helpful in an applied e-waste situation where the user needs to know

why a device is recommended for resale, repair or recycling. Valuation of the metals is integrated, further linking the classification results of images with economic decision support.

The framework does have a couple of drawbacks, however. First, only four iPhone models are included in the set and it does not generalize across any other generation of iPhone or Android versions, tablets, laptops, etc., without data collection and retraining. Second, there are a few hash groups that contain duplicate hashes in Stage-C, which introduces a small risk of leakage of the data—this should be resolved by full hash-based deduplication. Thirdly, the results are not the result of repeated experiments with multiple random seeds and confidence interval, but only single training runs. 4) The estimate of the metal value is an approximation and does not constitute a certified recycling quote. The actual value will vary depending on the efficiency of the recovery process, the internal condition of the item, labour costs, transport cost and market volatility.

The framework, however, shows that it is feasible in practice despite these drawbacks. The prototype of FastAPI and React illustrates that the pipeline can be wrapped as a web-accessible tool for the user to upload an image from their smartphone, and receive structured outputs such as model prediction, condition prediction, severity prediction (if applicable), estimated metal value, and recommendation.

VI. CONCLUSION

In this paper, the authors introduced a deep learning based five-stage framework for the classification of e-waste items, damage assessment, estimation of their metal value and recommendation for recycling. This system takes one image of a smartphone as input, classifies the image with a model, classifies the image with a condition, classifies the image with a conditional severity, estimates the value of the metal in the image and makes a recommendation using a rule. A custom dataset was created from the web and from real-world sources, while three model families are assessed: DenseNet121, YOLO11x-cls, and Vision Transformer-based models.

The final architecture pipeline selected had 91.91% test accuracy for smartphone model classification with DINOv2 ViT-B/14, 92.12% for normal/broken condition classification with Swin Transformer V2 Tiny, and 94.67% for minor/major damage severity classification with MaxViT-Tiny. The ViT-based models had better performance than the CNN and YOLO baselines on all three stages. Stage-D showcased hybrid MetalpriceAPI and fallback-based recyclable metal value estimation while Stage-E showed how this information can be used to provide practical recommendations: sell as is, repair, resell, sell for recycling or manual assessment.

Future works include rebuilding all the dataset splits using hash based deduplication, adding more iPhone generations and Android phones to the dataset, testing the system with external real world images, enhancing Stage-B broken class recall, multi-seed evaluation and using certified recycler pricing for

more accurate valuations. In conclusion, the proposed framework provides a real-world solution to the problem of supporting decision making in the field of e-waste management for smartphones and shows the feasibility of the deep learning predictions being linked to the real-world e-waste recycling and re-use workflow.

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